real-time volatility is often a key explanatory factor in these models, having improved volatility forecasts on hand will likely lead to more accurate models. And having more accurate models will allow investors to better react to changing market conditions. And most importantly, this ensures consistency between trading decisions and investing objectives of the fund.

Table 6.4 provides a comparison of the different volatility forecasting models.

Table 6.4 Volatility Forecasting Models

Volatility Model	Formula	Parameter(s)	Calculation
HMA	$\overline{\sigma}_t^2 = \frac{1}{N-1}\sum_{i=1}^{N} r_{t-i}^2$	n/a	By Definition
EWMA	$\widehat{\sigma}_t^2 = (1-\lambda)r_{t-1}^2 + \lambda\widehat{\sigma}_{t-1}^2$	λ	MLE
ARCH(1)	$\widehat{\sigma}_t^2 = \omega + \alpha r_{t-1}^2$	ω, α	OLS
GARCH(1,1)	$\widehat{\sigma}_t^2 = \omega + \alpha r_{t-1}^2 + \beta\widehat{\sigma}_{t-1}^2$	ω, α, β	MLE
VIX Adj.	$\widehat{\sigma}_t^2 = \overline{\sigma}_{t-1}^2 \cdot \left(\frac{VIX_{t-1}}{SP_{500}\overline{\sigma}_{t-1}}\right)^2$	n/a	By Definition
	$\widehat{\sigma}_t^2 = \beta_0 + \beta_1 \cdot VIX_{t-1}^2$	β_0, β_1	OLS

Problems Resulting from Relying on Historical Market Data for Covariance Calculations

Next we want to highlight two issues that may arise when relying on historical data for the calculation of covariance and correlation across stocks using historical price returns. These issues can have dire results on our estimates. They are:

- False Relationships.
- Degrees of Freedom.

False Relationships

It is possible for two stocks to move in the same direction and have a negative calculated covariance and correlation measure and it is possible for two stocks to move in opposite directions and have a positive calculated covariance and correlation measure. Reliance on market data to compute covariance or correlation between stocks can result in false measures.